TA14 | From Laboratory Efficacy to Authorized Deployment

Institutional Technical Examination | Version 3.0 | 15 October 2026

Independent TA14 technical governance examination framework for research-to-deployment decisions involving air-treatment technologies. Prepared for a potential written technical exchange; not reviewed, approved, funded or endorsed by EPA, Blueprint Biosecurity or CRADA participants.

Governing question

Does this proposed consequence have sufficient Admissible Evidence, Applicable Authority, and Established Standing to become reality NOW?

Decision boundaries

A laboratory performance result, an extrapolated field-performance claim, a health-benefit claim, and authorization to deploy are separate propositions. Each requires a separately bounded examination and disposition. Governance records do not substitute for independent scientific validation, competent safety review or regulatory approval.

Document control

Item	Record
Owner	TA14 Authority Governance Institution (independent draft)
Edition	v3.0 | institutional examination | 15 October 2026
Source baseline	TA14 v2.0 four-page institutional examination, 15 October 2026
Intended use	Nonconfidential technical discussion; no deployment authorization
Public reference	www.ta14exchange.com

Status: proposed examination methodology; not a finding concerning any specific device, research project, school, agency or CRADA.

01 | Institutional problem and scope

The research-to-deployment gap is not one problem. It combines scientific evidence, field representativeness, safety, operational continuity, institutional permission, communication and investment decisions. TA14 proposes to make the evidence-to-consequence boundary explicit, inspectable and repeatable.

Challenge raised in correspondence	Proposed TA14 contribution
Representative efficacy tests	Claim/endpoint register; protocol versioning; comparators, raw records and uncertainty examination.
Laboratory-to-field translation	Transferability matrix; unsupported inference flags; field verification and bounded claims.
Public confidence and support	Traceable claim language, limitations, accountable decision records and complaint/incident pathways.
Investment prioritization	Comparable readiness/gap records and decision conditions; not a substitute for economic evaluation.
Regulatory complexity	Jurisdiction and claim-specific issue map, authority/standing records and escalation routing.

What is and is not being proposed

TA14 could examine the integrity of a decision pathway, establish a governance schema and operate versioned accountability records. It does not certify efficacy, grant regulatory approval, assume school-site authority, decide clinical risk or independently authorize deployment.

Nonconfidential first exchange

Select one representative technology and one narrowly specified proposed consequence. Request only the documents the owner is permitted to share. Define a written scope, named technical contacts and review rights before any nonpublic disclosure.

02 | Eight-anchor execution architecture

Anchor	Required proof / examination output
Reality	Specify observed conditions and relevant physical context; do not equate a surrogate with a clinical outcome.
Record	Preserve original observations, source identifiers, protocol and instrument provenance.
Continuity	Verify time, configuration, custody, calibration, changes and comparison windows.
Admissibility	Test evidence relevance, completeness, uncertainty and suitability for this precise claim.
Binding	Bind evidence to a specific actor, site, action, configuration, timeframe and claim.
Commit	Record a competent actor's disposition, authority, standing, constraints and review trigger.
Execution	Permit only the bounded action actually authorized; HOLD is not permission.
Outcome	Record field observations, deviations, adverse events, stop events and reassessment.

Non-substitution rule

Observed is not inferred. Inferred is not authorized. Capability does not equal authority. A laboratory observation may be admissible for a narrow laboratory claim while being insufficient for a classroom health-benefit claim or an occupied-site deployment decision.

Architecture failure examples

Missing calibration undermines Record/Admissibility; changed equipment without logs undermines Continuity; absent school authorization defeats Commit/Execution; missing stop criteria prevents an acceptable execution boundary. These failures should be recorded separately, not hidden in a single readiness score.

03 | Evidence package and provenance

Evidence family	Minimum record
Claim and endpoint	Exact proposition, measured endpoint, intended population, exposure context and limits.
Test identity	Device/model/configuration; protocol/version; laboratory setup; challenge agent; comparator/control.
Measurement integrity	Raw time-stamped data, instrument identity, calibration, detection limits, uncertainty, exclusions.
Transfer conditions	Room size, airflow, mixing, sources, occupancy, environmental and operational variability.
Safety	Potential byproducts, transformation products, secondary exposure, failures and competent review.
Decision rights	Named actor, capacity, standing basis, authority relationship, permits and constraints.
Operations	Installation, duty cycle, maintenance, changes, power, monitoring, stop and rollback.
Outcome	Baseline, acceptance criteria, post-action measurements, incidents, complaints and reassessment.

Raw-versus-adjusted evidence rule

Preserve raw observations as the primary record. Store adjusted/normalized records as separately linked derivatives with method, input data, software/version, rationale, adjustment magnitude and reviewer. Never silently overwrite or replace the original record.

Provenance and integrity

Use persistent record IDs, timestamps, custody/transfer history and hashes where feasible. A hash helps detect alteration; it does not prove scientific validity, accurate calibration or rightful collection. Record uncertainty and missing evidence explicitly.

04 | Scientific claim examination

This is an examination of what the available scientific record can support, not a replacement for subject-matter scientific peer review.

Examination dimension	Question / expected evidence
Endpoint validity	What was measured, by what method, and is that endpoint the same as the proposed claim?
Protocol suitability	Are challenge conditions, replicates, controls, sampling windows and exclusions described?
Measurement quality	What calibration, uncertainty, limits of detection and data-integrity records exist?
Reproducibility	Are repeatability, inter-run variability and independent replication described or missing?
Causal attribution	Could ventilation changes, baseline drift, occupancy or co-interventions explain results?
Claim language	Does wording stay inside tested conditions, population and endpoint?
Independent review	Which scientific/safety questions require competent external review before decision?

Disposition principle

An acceptable laboratory finding can support a bounded statement such as a reported surrogate reduction under specified chamber conditions. It cannot by itself establish infection-risk reduction, occupant benefit, field efficacy or deployment permission.

No manufactured thresholds

No universal numerical efficacy threshold, uncertainty tolerance or field-transfer factor is supplied here. Such criteria must be defined and justified for the technology, claim, intended use and applicable scientific/regulatory context.

05 | Laboratory-to-field transferability

Dimension	Laboratory evidence	Field examination
Endpoint	Surrogate and test duration	Relevant target endpoint; evidence bridge to claim
Air environment	Volume, mixing, ventilation, T/RH	Site airflow, gradients, variable operation
Sources/people	Challenge dose and test occupancy	Real source rates, occupancy, susceptible groups
Device	Settings, position, duty cycle	Installed configuration, maintenance, changes
Measurement	Replicates, controls, error	Baseline, confounders, comparability, drift
Safety	Lab hazards and byproducts	Exposure pathways and site-specific safety review

Transfer test sequence

1. Identify the exact laboratory proposition. 2. Specify the proposed field proposition. 3. Map differences and uncertainty by dimension. 4. Identify evidence-backed bridges, not assumptions. 5. Determine which claims remain unsupported. 6. Define independent field verification, acceptance criteria and reassessment conditions.

Decision logic

Material uncharacterized mismatch produces HOLD for extrapolated claims. A site-specific prohibition or absent legal decision right may separately produce DENY for deployment. Competent specialist review may require ESCALATE; none of these outcomes automatically invalidates a properly bounded laboratory observation.

06 | Six operational examination gates

Gate	Required record	Failure / route
G1 Claim boundary	Claim, endpoint, protocol and evidence map	HOLD unsupported scope expansion
G2 Environmental match	Lab-field matrix, site survey, comparability rule	HOLD material unknown mismatch
G3 Safety and exposure	Hazard/byproduct assessment and specialist owner	ESCALATE unresolved safety judgment
G4 Operational continuity	Configuration, calibration, service and duty-cycle logs	HOLD unverified continuity
G5 Authority and standing	Permission, named actor, ASA anchors, jurisdiction review	DENY unauthorized action
G6 Verification and exit	Baseline, acceptance, monitoring, stop/rollback owners	HOLD absent bounded controls

Gate ownership

Scientific reviewers evaluate test claims; safety specialists evaluate relevant exposure hazards; site authorities decide local permissions; operators establish operating controls. TA14 can document and route those responsibilities but cannot assume them by generating a gate report.

Gate completion record

For each gate: status; evidence IDs; reviewer identity and capacity; findings; uncertainty; corrective action; due/review date; authority required to close; resulting disposition. A gate marked complete does not supersede a prohibition or authorize a different consequence.

07 | ASA standing and authority boundary

ASA anchor	Required question
ACTOR	Who is proposing, reviewing, approving or performing the action?
CAPACITY	In which organizational or professional role is that actor acting?
STANDING BASIS	What documentary basis permits participation or decision?
AUTHORITY RELATIONSHIP	Who delegated or conferred the right, for which scope and jurisdiction?

Illustrative role separation

A manufacturer may present laboratory evidence; a researcher may assess scientific validity; a qualified safety specialist may evaluate hazards; a school decision maker may authorize site activity; a regulator may hold jurisdiction-specific authority. No actor gains another actor's rights merely by participating in the same project.

Regulatory issue spotting

Identify potentially relevant EPA, FDA, OSHA, state, local, education and occupational interfaces based on the technology, intended claim and setting. This is a screening map only: applicability, exemptions, approvals and jurisdiction require competent determination. Do not imply that any named agency has reviewed this TA14 document.

Federation boundary

Authority context and evidence may be transmitted between institutions; execution authority must be established locally for the receiving site and proposed consequence. A research agreement is not automatically a site deployment authorization.

08 | Human-consequence and stop boundary

Concern	Examination evidence / safeguard
Byproducts	Known or plausible transformation products, exposure pathways and relevant tests.
Susceptible occupants	Site population and potentially elevated exposure vulnerability; competent review.
Operational failure	Loss of power, filter/media failure, incorrect settings, unintended chemical generation.
Occupancy and consent	Relevant site policy, notices, access, operating hours and permission.
Complaints/incidents	Reporting channel, triage owner, retention, escalation and investigation trail.
Stop/rollback	Trigger, immediate action, named authorized owner and restoration verification.
Outcome	Pre/post comparability, adverse events, deviations and unresolved uncertainty.

Stop conditions are not decorative

The proposed deployment plan must specify observable conditions requiring pause, shutdown, specialist review or reversal. Monitoring data alone do not confer authority to override site safety protocols. Emergency and regulatory obligations remain controlling.

Health-claim boundary

A measured surrogate concentration reduction is not equivalent to an established infection-risk or clinical-health outcome. Any claim of health benefit requires an appropriately justified evidentiary bridge and relevant competent review.

09 | Worked hypothetical classroom examination

Fictional teaching case only. A manufacturer reports reduced airborne surrogate concentration in a controlled chamber. A school considers installation in occupied classrooms and wishes to claim proven infection-risk reduction. No real device, school, or CRADA result is being evaluated.

Proposal	Observed / missing	Disposition
A: Publish narrow laboratory result	Chamber report described; raw records, protocol and uncertainty still to be examined	HOLD pending source review; potentially ALLOW bounded claim
B: Claim classroom infection-risk reduction	No validated field/exposure-to-health bridge in supplied scenario	HOLD broad claim
C: Install and operate in occupied room	Missing site survey, byproduct review, operating controls and signed permission	HOLD; DENY if action is unauthorized

Evidence register - illustrative

LAB-001 chamber report (reported, not validated); RAW-001 raw files (not supplied); SITE-001 airflow survey (missing); SAFE-001 byproduct review (missing); OPS-001 duty-cycle/maintenance/stop plan (missing); AUTH-001 signed site authorization (missing); BASE-001 field baseline (missing).

Reopen conditions

Obtain and independently review laboratory records; bound each claim; characterize the site and baseline; complete competent safety review; verify installation and operating plan; establish documented local authority and standing; define monitoring, acceptance, stop, rollback and review triggers.

10 | Decision record and audit schema

Field	Illustrative record value
Proposal/version	AIR-DEPLOY-EXAMPLE-001 / v0.3
Site / timeframe	Hypothetical occupied classroom / date TBD
Proposed consequence	Operate specified device during occupancy
Evidence binding	LAB-001; RAW-001 missing; SITE-001 missing; SAFE-001 missing
Authority / ASA	School decision maker to be named; standing and authority unverified
Disposition	HOLD - site transfer, safety, operations and permission incomplete
Conditions	No deployment under this record; complete gates G2-G6
Reopen trigger	New evidence and local authorization reviewed by competent actors
Stop / rollback owner	Must be assigned before any future authorized operation
Outcome links	BASE-001, POST-001, INCIDENT-001 (prospective records)

Decision separation

Store separate decision objects for laboratory claim publication, field-performance claim, health-benefit claim and physical deployment. Each has its own evidence binding, authority/standing test, disposition, signatory and expiration/review trigger.

Auditability

Maintain immutable source references where feasible, versioned amendments, recorded reasons for changing dispositions, signatory authority, and linked post-intervention outcomes. Do not represent the sample IDs above as actual collected evidence.

11 | Research-to-deployment interface

The proposed interface is a governance integration pattern, not an existing EPA or Blueprint system and not evidence that TA14 software is deployed.

Input layer	Governed transition	Output / owner
Laboratory protocols, results, raw records	Identity, provenance, uncertainty and bounded claim mapping	Scientific claim examination / qualified reviewers
Site surveys, baselines and operating plans	Transferability and continuity assessment	Field evidence package / site technical team
Safety findings and jurisdiction issues	Specialist escalation and authority verification	Review determinations / competent authorities
Actor and permission records	ASA standing and local execution boundary	Authorization record / site decision maker
Live operational and outcome records	Monitoring, stop events, reassessment and audit links	Post-intervention record / accountable operator

Proposed technical data contract

Each proposal would reference unique claim ID, site ID, configuration ID, evidence object IDs, time validity, provenance, uncertainty, ASA actor/role/authority record, gate results, disposition, constraints, review trigger and outcome links. Interface and security specifications would require separate design and approval.

No authority transfer by API

Data interoperability is not legal delegation. A receiving institution must independently establish applicable authority and standing before local execution.

12 | Engagement lanes, questions and limitations

Lane	Potential bounded deliverable
EXAMINE	Nonconfidential evidence-gap and authority-boundary matrix for one specified use case.
ESTABLISH	Versioned evidence schemas, ASA standing records, gate definitions and decision templates.
OPERATE	Ongoing continuity, outcome, reassessment and audit record management, subject to agreement.

Written questions for a technical counterpart

What exact laboratory claim and endpoint are being evaluated? Which real-world setting and consequence are contemplated? Which transferability gaps are unresolved? Who owns scientific review, safety review, site permission and regulatory interpretation? What monitoring and stop criteria already exist? Which records may be shared nonconfidentially?

Conditions for further work

A written scope should address deliverables, technical review rights, confidentiality, intellectual property, data access, security, liability, licensing and compensation before confidential exchange or implementation. These are proposed capabilities, not funded work, procurement commitments or CRADA participation.

Independence and examination limits

This document is not a scientific validation study, clinical risk assessment, engineering design, regulatory ruling, legal opinion or authorization to deploy. It contains no asserted empirical findings. EPA, Blueprint Biosecurity and named researchers have not endorsed it. Decisions remain with actors possessing applicable authority and established standing.

Showroom teaching sequence

01 Laboratory-to-reality gap; 02 Tested claim boundary; 03 Provenance and baseline continuity; 04 Transferability and six gates; 05 Human consequence; 06 Authority and standing; 07 Four dispositions and separate consequences; 08 Authorized execution and verified outcome.